

SMART DONATION CONNECT

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BONAFIDE CERTIFICATE

Certified that this project report "SMART DONATION CONNECT-An AI- Enable Platform For Reducing Waste And Connecting Donors With people In Need",is the Bonafide work of MANASHA HARSHINI-24619121, KIRTHANA.K-24619117, BHUVANESWARI.M-24619105 carried out the project under my supervision. Certified further, that to the best of my knowledge the work reported here does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate

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ABSTRACT

Abstract— Across millions of people in India are struggling to meet their daily needs even though surplus food, clothing, electronic devices, and other resources are available. Hotels, restaurants, event halls, businesses, and households often waste these resources, while orphanages, old-age homes, NGOs, poor families, and people in remote villages face shortages. This project proposes Smart Donation Connect, an AI-enabled web application that connects verified donors with verified beneficiaries across the country. The platform allows individuals, businesses, and organizations to donate surplus resources, while beneficiaries can register their requirements through a secure verification process. An AI-based food demand prediction model analyzes historical sales data, seasonal trends, festivals, weather conditions, and customer behavior to estimate food demand and potential surplus. An AI-powered recommendation system matches donors with suitable beneficiaries based on location, urgency, donation type, and requirements. Real-time notifications, verified user profiles, and transparent donation tracking help improve trust and efficiency. The proposed system aims to reduce resource wastage, encourage charitable participation, and promote social welfare and sustainable development.

Keywords— Artificial Intelligence, Web Development, Food Waste Prediction, Smart Donation Platform, Donation Matching, Resource Sharing, NGOs, Social Welfare, Real-Time Notifications, Sustainable Development.

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Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
NGO	Non- Government Organization
UI	User Interface
UX	User Experience
DBMS	Database Management System
SQL	Structured Query Language
API	Application Programming Interface
GPS	Global positioning System
OTP	One-time Password
CNN	Convolutional Neural Network
IOT	Internet of Things

Abbreviation	Full Form
HTML	Hypertext Markup Language
CSS	Cascading Style Sheets
JS	Java Script
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
JSON	JavaScript Object Notation
DFD	Data Flow Diagram
SDG	Sustainable Development Goals
CRUD	Create,Read, Update, Delete
URl	Uniform resource location

CHAPTER 1.

1. INTRODUCTION

Large quantities of food, clothing, electronic devices, educational materials, and household resources are wasted even though many people and organizations require these resources. Restaurants, hotels, educational institutions, households, and businesses may have surplus resources that could help others. However, without an organized platform, it is often difficult to identify genuine recipients and distribute available resources efficiently.

Traditional donation systems generally depend on manual searching, personal contacts, or social media communication. These approaches may not efficiently identify suitable recipients based on their actual needs. In addition, donors may not have sufficient information about the quantity, urgency, location, and condition of the resources required by receivers.

To address these challenges, this project proposes Smart Donation Connect, an AI-enabled platform that connects donors with verified receivers. The system uses intelligent matching techniques to consider resource type, quantity, condition, urgency, and geographical location. In addition, the platform can analyze historical food-related information to predict future demand and identify possible surplus resources.

The proposed system aims to reduce resource wastage while improving the efficiency and reliability of the donation process. It provides a structured platform through which donors can list available resources and receivers can post their requirements. The system then assists in identifying suitable donor–receiver pairs and tracks the donation process until completion.

1.1 OVERVIEW OF THE PROJECT

Smart Donation Connect is an AI-enabled web application designed to reduce resource wastage and connect donors with people and organizations in need. Large quantities of food, clothing, electronic devices, educational materials, and other reusable items are often wasted by hotels, restaurants, businesses, and households. At the same time, NGOs, orphanages, old-age homes, educational institutions, and economically disadvantaged communities face shortages of essential resources.

The proposed system provides a common platform where donors can register and list available resources, while beneficiaries can register and post their requirements. A verification process helps improve the reliability of donor and beneficiary profiles. The system uses intelligent donor–receiver matching to identify suitable recipients based on donation type, quantity, urgency, availability, and geographical location.

An important feature of the project is the AI-based food demand prediction model. It analyzes historical sales data, seasonal trends, festivals, weather conditions, and customer patterns to estimate food demand and identify potential surplus food. This information can help restaurants and hotels plan food preparation and reduce unnecessary waste.

The platform also provides real-time notifications and donation tracking to support the donation process. Donations are tracked through different stages, including posting, matching, acceptance, collection, delivery, and completion. These features help users monitor the progress of donations and improve coordination between donors and beneficiaries.

1.2. OBJECTIVE

The main objective of Smart Donation Connect is to reduce resource wastage, improve donation efficiency, and ensure that available resources reach genuine beneficiaries. By combining artificial intelligence, location-based matching, user verification, and donation tracking, the proposed system aims to encourage charitable participation and support social welfare and sustainable development.

2. LITERATURE REVIEW

2.1 Literature Overview

The literature review examines existing approaches related to food waste reduction, food demand prediction, online donation systems, and resource sharing. These approaches demonstrate how artificial intelligence, machine learning, and digital platforms can help reduce resource wastage and improve the distribution of essential items to people in need.

Machine-learning-based food demand forecasting helps estimate future food requirements by analyzing historical data and relevant patterns. This can support hotels and restaurants in planning food preparation and identifying possible surplus food. Digital food-sharing and online donation platforms provide methods for connecting people who have extra resources with individuals or organizations that require assistance.

Location-based resource-sharing systems can help identify nearby recipients and improve the coo rdination of donations. However, existing approaches may focus mainly on food sharing or may not combine demand prediction with intelligent donor–receiver matching. Therefore, an integrated platform is needed to manage different donation categories, verify users, identify suitable beneficiaries, and track donations efficiently.

Smart Donation Connect addresses these requirements by combining AI-based food demand prediction, donor and receiver verification, intelligent matching, location-based recommendations, real-time notifications, and donation tracking within a single platform.

2.2 Literature Survey

1. Smart Food-Sharing Platforms for Social Sustainability: A Heuristic Algorithm Approach (2026)

This work focuses on food-sharing platforms and their role in supporting social sustainability. Fo od-sharing approaches help connect available food resources with people who need them. Such systems can contribute to reducing food waste and improving resource utilization. Smart Donation Connect extends this concept by supporting multiple types of donations and combining food demand prediction with donor–receiver matching.

2. Why Do Hospitality and Food Service Businesses Use Food Sharing Platforms? (2026)

This study examines the use of food-sharing platforms by hospitality and food-service businesses. Hotels and restaurants may have surplus food that could be distributed to people in need instead of being wasted. Digital platforms can support the sharing process by connecting available food with potential recipients. Smart Donation Connect builds on this concept by including location-based matching, user verification, notifications, and donation tracking.

3. Machine Learning-Based Demand Forecasting for Bakery Food Waste Reduction (2024)

This work relates to the use of machine learning for forecasting food demand and reducing bakery food waste. Demand forecasting can help businesses estimate the quantity of food required and identify potential surplus. Smart Donation Connect applies the idea of food demand prediction to its proposed platform, using historical food-related information and relevant patterns to estimate demand and possible surplus for donation.

Research Gap

The reviewed approaches address food sharing, demand forecasting, and resource distribution. However, these approaches may focus on individual aspects rather than combining all the required functions in one platform. Smart Donation Connect proposes an integrated approach that supports different donation categories, verified users, AI-based food demand prediction, intelligent donor–receiver matching, location-based recommendations, real-time notifications, and donation tracking. This approach aims to improve donation coordination, reduce resource wastage, and help available resources reach genuine beneficiaries.

CHAPTER 3

SYSTEM DESIGN

3.1 EXISTING SYSTEM

Existing donation systems mainly depend on manual searching, personal contacts, and social media communication to connect donors with people in need. Some digital platforms focus on food sharing, while others support specific types of donations. These systems may not consider all the important factors required for efficient resource distribution.

Identifying genuine beneficiaries and matching them with suitable donors can be difficult. Location, quantity, urgency, resource condition, and availability may not be considered together. In addition, the lack of integrated food demand prediction and donation tracking can make the process less organized.

Disadvantages of the Existing System

- Manual searching requires more time.
- Verification of genuine donors and beneficiaries may be limited.
- Many platforms focus on a single type of donation.
- Matching donations with actual requirements can be difficult.
- Tracking the complete donation process may be challenging.
- Food surplus may not be predicted efficiently.

3.2 PROPOSED SYSTEM

Smart Donation Connect is an AI-enabled web platform designed to connect verified donors with genuine beneficiaries. It supports donations such as food, clothing, electronic devices, educational materials, household items, and financial assistance.

The system provides separate registration for donors and beneficiaries. Donors can list available resources with details such as quantity, condition, location, and availability. Beneficiaries can post their requirements by specifying the resource type, quantity, urgency, and location.

An AI-based food demand prediction model analyzes historical sales data, seasonal trends, festivals, weather conditions, and customer patterns to estimate food demand and potential surplus. The intelligent matching module identifies suitable donors and beneficiaries based on location, donation category, urgency, quantity, and availability.

The platform also provides real-time notifications and donation tracking. The donation status is monitored from posting and matching to acceptance, collection, delivery, and completion.

Advantages of the Proposed System

- Connects donors with verified beneficiaries.
- Supports multiple categories of donations.
- Uses AI-based food demand prediction.
- Provides intelligent and location-based matching.
- Sends notifications when suitable resources are available.
- Tracks donations throughout the process.
- Helps reduce resource wastage and improve donation coordination.

3.3 SYSTEM ARCHITECTURE

The system architecture describes the main components of Smart Donation Connect and how they work together.

The donor and beneficiary first register on the platform and complete the verification process. Donors post available resources, while beneficiaries submit their requirements. The AI-based food demand prediction module estimates potential food surplus using historical information. The intelligent matching module compares available donations with beneficiary requirements based on location, category, urgency, quantity, and availability.

When a suitable match is identified, the system sends notifications to the relevant users. The donation is then accepted, collected, and delivered. Finally, the system updates the donation status until completion.

System Architecture Flow

User Registration → Verification → Donation Listing / Requirement Posting → AI Food Demand Prediction → Intelligent Matching → Notification → Donation Acceptance → Collection → Delivery → Tracking and Completion

3.4 HARDWARE REQUIREMENTS

The following are suggested basic hardware requirements for developing and using the proposed web application. The source paper does not specify exact hardware models or minimum specifications.

- Computer or laptop
- Processor suitable for web development and machine-learning tasks
- RAM sufficient to run the development tools
- Storage for project files and historical data
- Internet connection
- Smartphone or computer for accessing the web application

CHAPTER 4

IMPLEMENTATION

4.1 METHODOLOGY

The proposed system, Smart Donation Connect, is an AI-enabled platform designed to connect donors with people and organizations in need. The methodology involves user registration, verification, donation listing, requirement posting, AI-based food demand prediction, intelligent matching, notifications, and donation tracking.

Initially, donors and beneficiaries register on the platform and complete the verification process. Donors provide information about the available resources, while beneficiaries specify their requirements. The system processes these details to identify suitable donor–beneficiary matches.

The food demand prediction module analyzes historical sales data, seasonal trends, festivals, weather conditions, and customer patterns to estimate food demand and potential surplus. The intelligent matching module considers donation category, location, quantity, urgency, and availability to identify suitable matches.

Finally, notifications are sent to the relevant users, and the donation status is tracked from posting to completion. This methodology aims to improve donation coordination, reduce resource wastage, and help available resources reach genuine beneficiaries.

4.2 IMPLEMENTATION

The implementation of Smart Donation Connect consists of different modules that work together to support donation management and resource distribution. Each module performs a specific function, from managing user information to tracking completed donations.

4.2.1 USER REGISTRATION AND VERIFICATION MODULE

This module allows donors and beneficiaries to register on the platform. Users provide the necessary information to create their profiles. The verification process helps improve the reliability of registered users and supports trustworthy donation activities.

4.2.2 DONATION LISTING MODULE

The donation listing module allows donors to list available resources, including food, clothing, electronic devices, educational materials, household items, and financial assistance. Donors provide relevant details such as resource category, quantity, condition, location, and availability.

This information helps the system identify resources that may satisfy the requirements of beneficiaries.

4.2.3 BENEFICIARY REQUIREMENT MODULE

This module allows beneficiaries to submit their requirements through the platform. They can specify the type of resource needed, required quantity, urgency, and location.

The submitted information helps the system understand the needs of beneficiaries and compare them with available donations.

4.2.4 AI-BASED FOOD DEMAND PREDICTION MODULE

The food demand prediction module uses historical sales data, seasonal trends, festivals, weather conditions, and customer patterns to estimate future food demand and potential surplus.

The predictions can help food providers understand expected demand and identify possible surplus food. This supports better food planning and helps reduce unnecessary wastage.

4.2.5 INTELLIGENT DONOR–BENEFICIARY MATCHING MODULE

This module identifies suitable matches between available donations and beneficiary requirements. It considers factors such as donation category, geographical location, quantity, urgency, and availability.

The matching process helps connect suitable donors with beneficiaries who require the available resources. It aims to improve the efficiency of donation distribution.

4.2.6 NOTIFICATION AND DONATION TRACKING MODULE

The notification module informs relevant users when suitable donation opportunities or matches are available. This helps donors and beneficiaries coordinate the donation process.

The tracking module monitors donation progress through stages such as posting, matching, acceptance, collection, delivery, and completion. It provides a structured way to follow the donation process and supports better coordination between users.

IMPLEMENTATION SUMMARY

Smart Donation Connect integrates user registration, verification, donation listing, beneficiary requirements, food demand prediction, intelligent matching

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 CONCLUSION

Smart Donation Connect prese nts an AI-enabled approach to reducing resource wastage and connecting donors with people and organizations in need. The proposed system combines user registration and verific ation, multiple donation categories, AI-based food demand prediction, intelligent donor–receiver matching, location-based matching, real-time notifications, and donation tracking.

The platform allows donors to list available resources such as food, clothing, electronic devices, educational materials, household items, and financial assistance. Beneficiaries can post their requirements, ena bling the system to identify suitable matches based on location, donation category, urgenc y, quantity, and availability.

The AI-based food demand prediction component uses historical sales data, seasonal trends, festivals, weather conditions, and customer patterns to estimate food demand and potential surplus. The matching and tracking modules support better coordination and help monitor donations from posting to completion.

Overall, Smart Donation Connect provides an integrated approach to improving donation management, reducing manual searching, and helping available resources reach genuine beneficiaries. The proposed platform can support social welfare, encourage charitable participation, and contribute to sustainable resource utilization.

5.2 FUTURE WORK

The proposed system can be improved further through testing with real-world datasets. The accuracy of food demand prediction and the performance of the donor–receiver matching process need to be evaluated using actual data.

Future work can focus on improving the machine-learning model using reliable historical food data and relevant factors such as seasonal trends, festivals, weather conditions, and customer patterns. The matching process can also be evaluated to determine how effectively it connects available resources with beneficiary requirements.

Further improvements can include enhancing location-based matching, improving notification response, and strengthening donation tracking. These developments can help make the platform more efficient and reliable.

The system can also be evaluated using measurable parameters such as prediction accuracy, matching performance, response time, and reduction in resource wastage. Such evaluation will help determine the effectiveness of Smart Donation Connect in practical donation-management scenarios.

5.3 SOURCE CODE

The source code of Smart Donation Connect is designed to support the main functions of the proposed donation management platform. These functions include user registration and verification, donation listing, beneficiary require ment posting, AI-based food demand prediction, intelligent donor–receiver matching, real-time notifications, and donation tracking.

The code implementation should integrate these modules to provide a structured donation process. Donors can list available resources, while beneficiaries can submit their requirements. The matching module identifies suitable donation opportunities based on location, donation category, urgency, quantity, and availability.

The AI-based food demand prediction module analyzes historical data and relevant factors to estimate food demand and potential surplus. The notification and tracking modules help users follow the progress of donations from pos ting to completion.

Note: The source paper describes the proposed system but does not provide the actual program code or specify the programming language used.

5.4 RESULTS

Smart Donation Connect presents an integrated approach to donation management by combining user verification, resource sharing, food demand prediction, intelligent matching, notifications, and donation tracking.

The proposed system supports multiple donation categories, including food, clothing, electronic devices, educational materials, household items, and financial assistance. It is designed to connect donors with suitable beneficiaries according to their requirements and geographical locations.

The AI-based food demand prediction component aims to estimate food demand and identify potential surplus resources. The intelligent matching and tracking features are intended to improve coordination between donors and beneficiaries and support the donation process from posting to completion.

The proposed architecture demonstrates how these components can work together within a single platform. However, actual prediction accuracy, matching performance, response time, and reduction in resource wastage require further evaluation using real-world datasets.

Overall, Smart Donation Connect provides a proposed solution for improving donation coordination, reducing resource wastage, and supporting social welfare.

5.4 Result

Output



Figure-1 logo

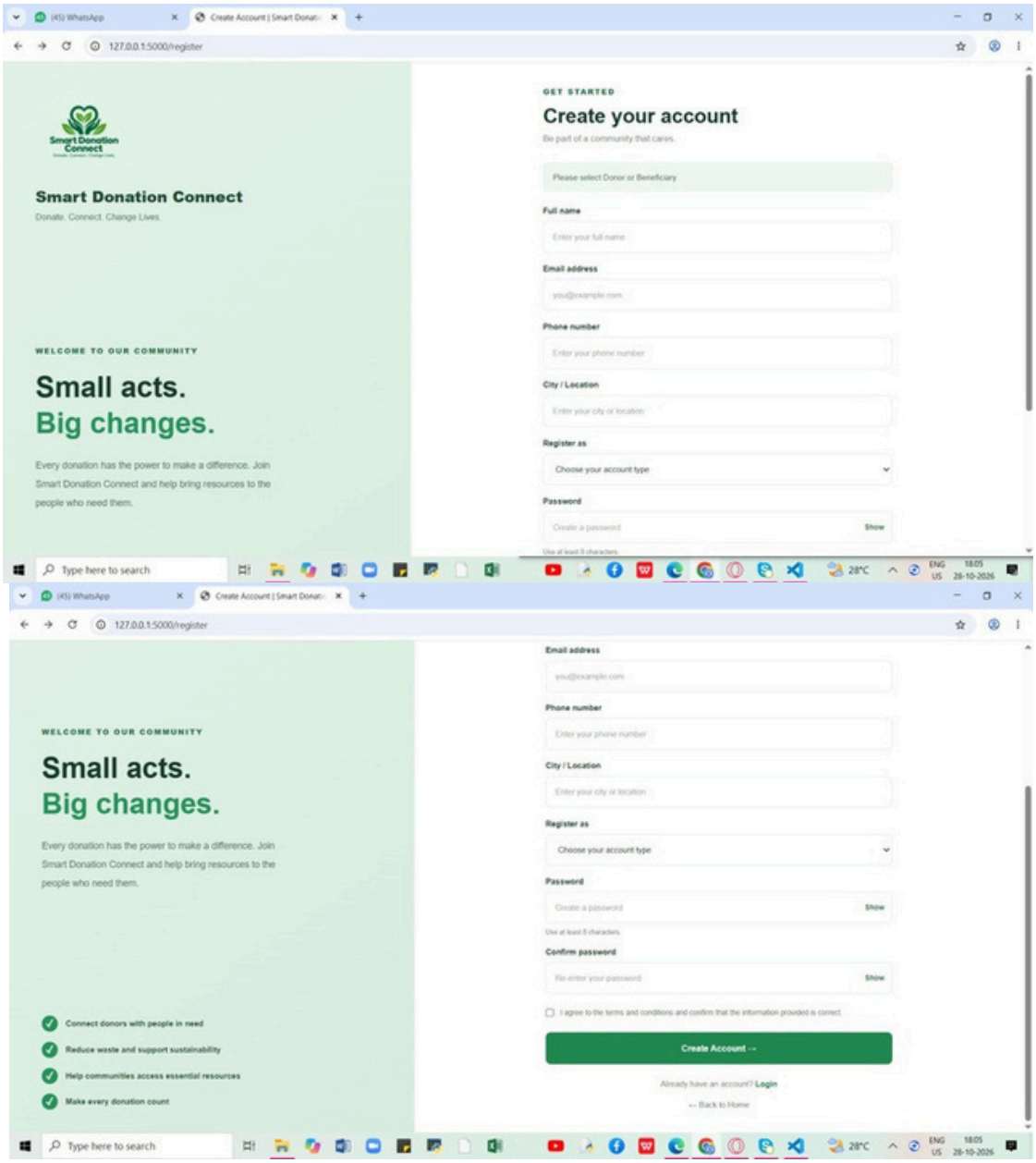


Figure-2 Registration page

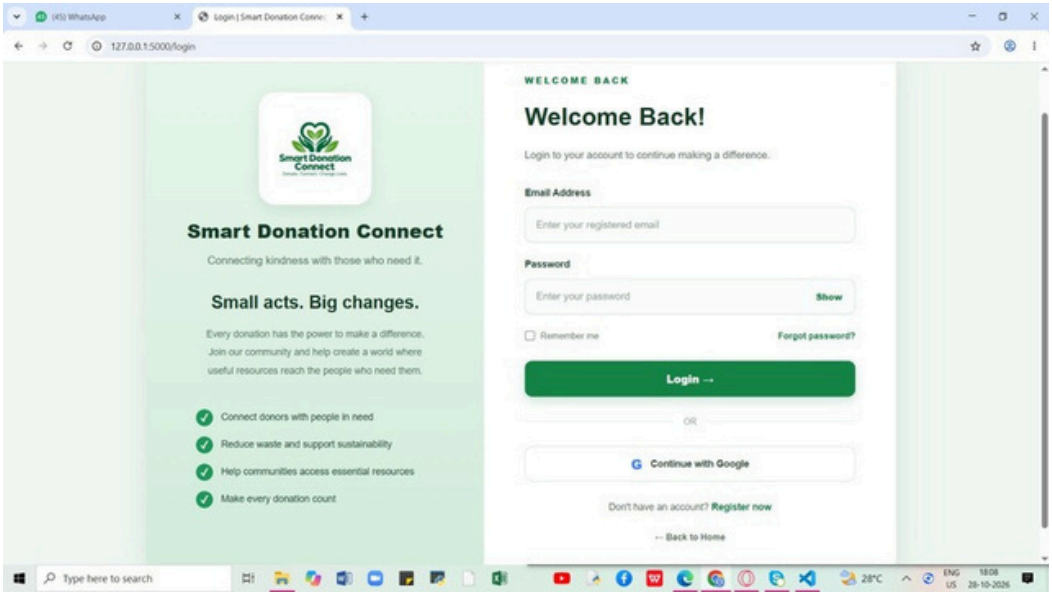
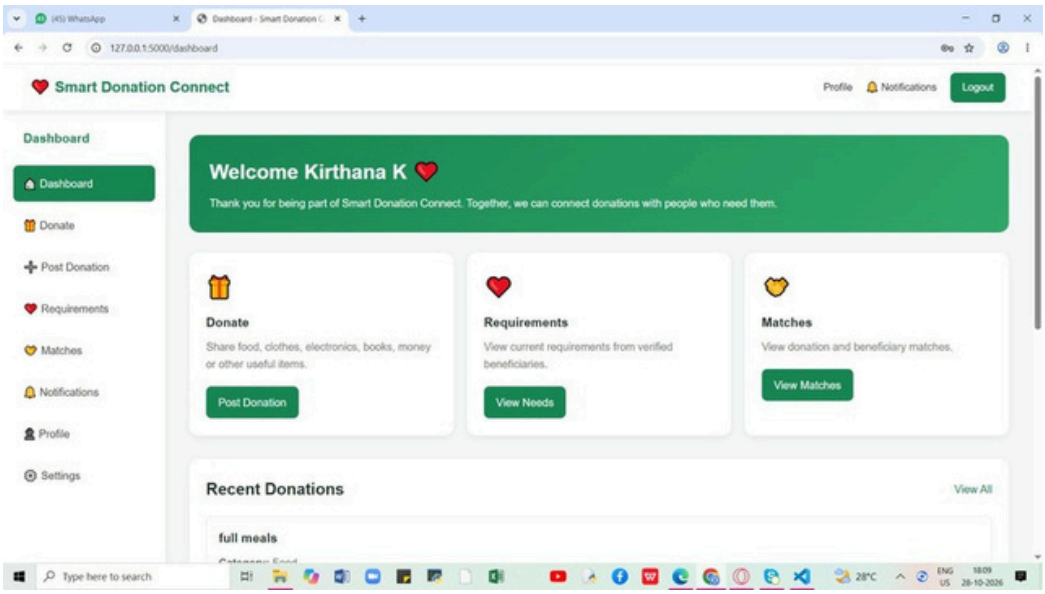


Figure-3 Login page



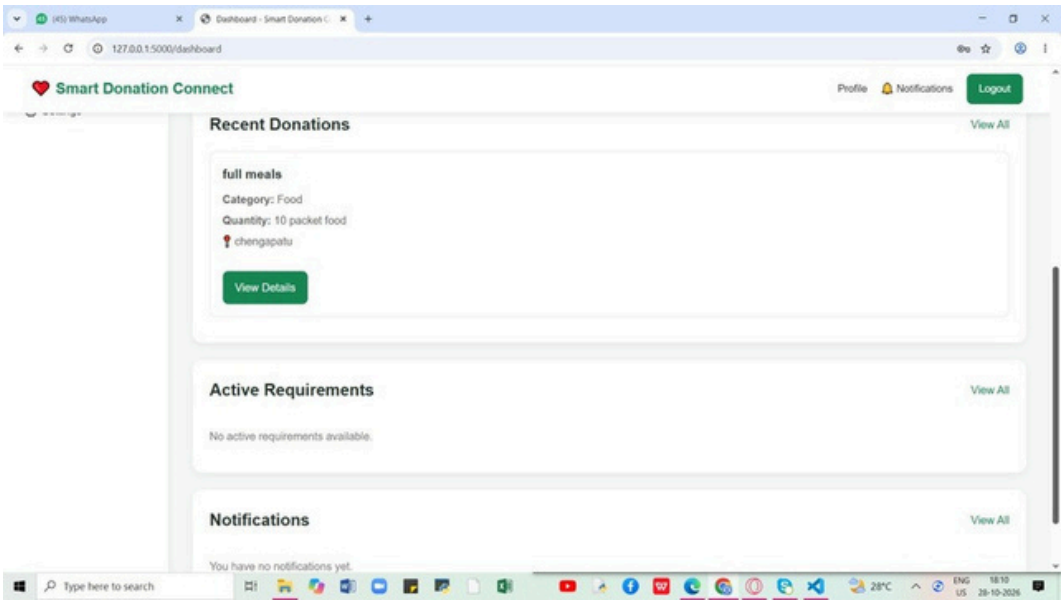


Figure-4 Main page

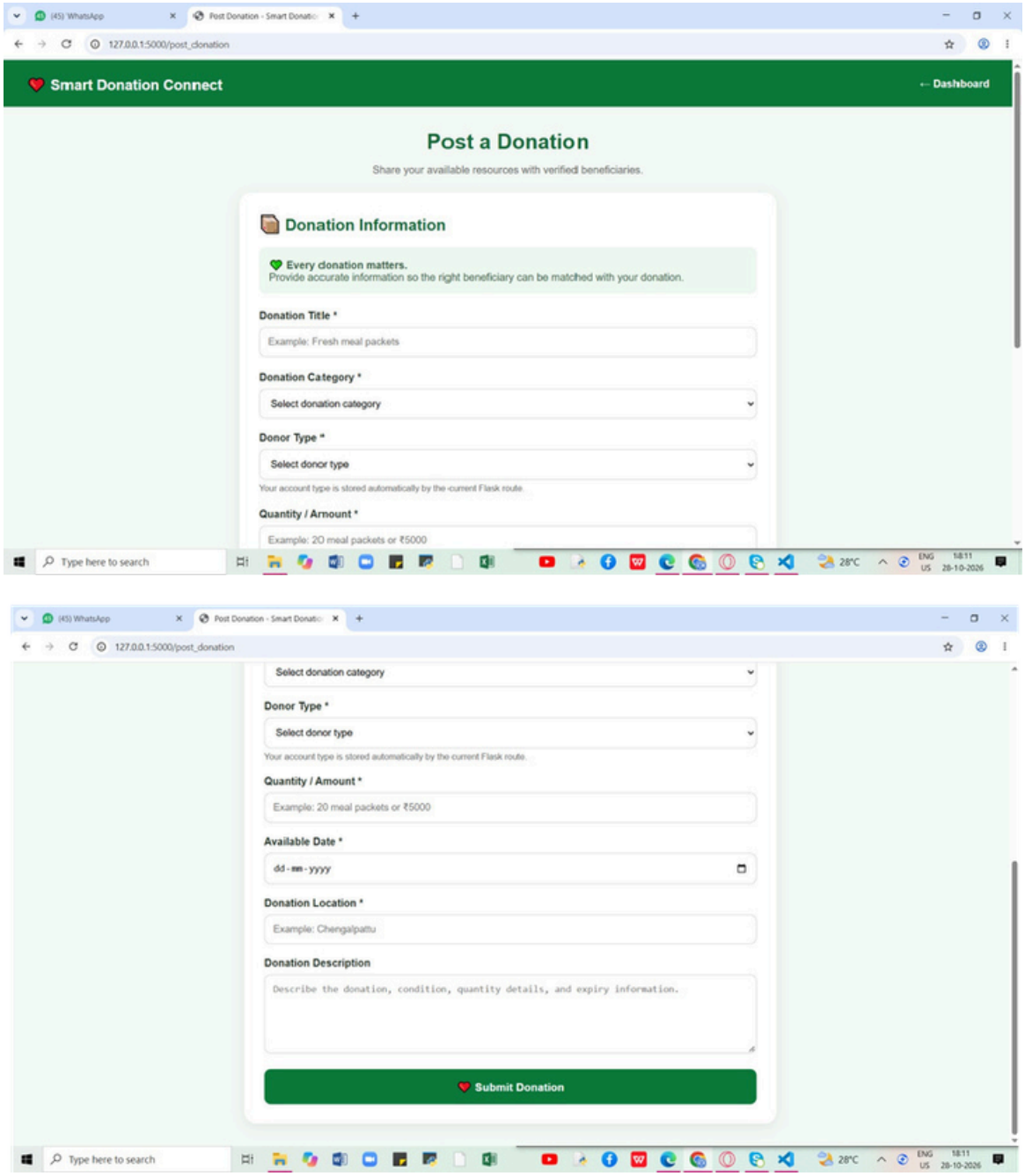


Figure-5 Donate and post donation dashboard

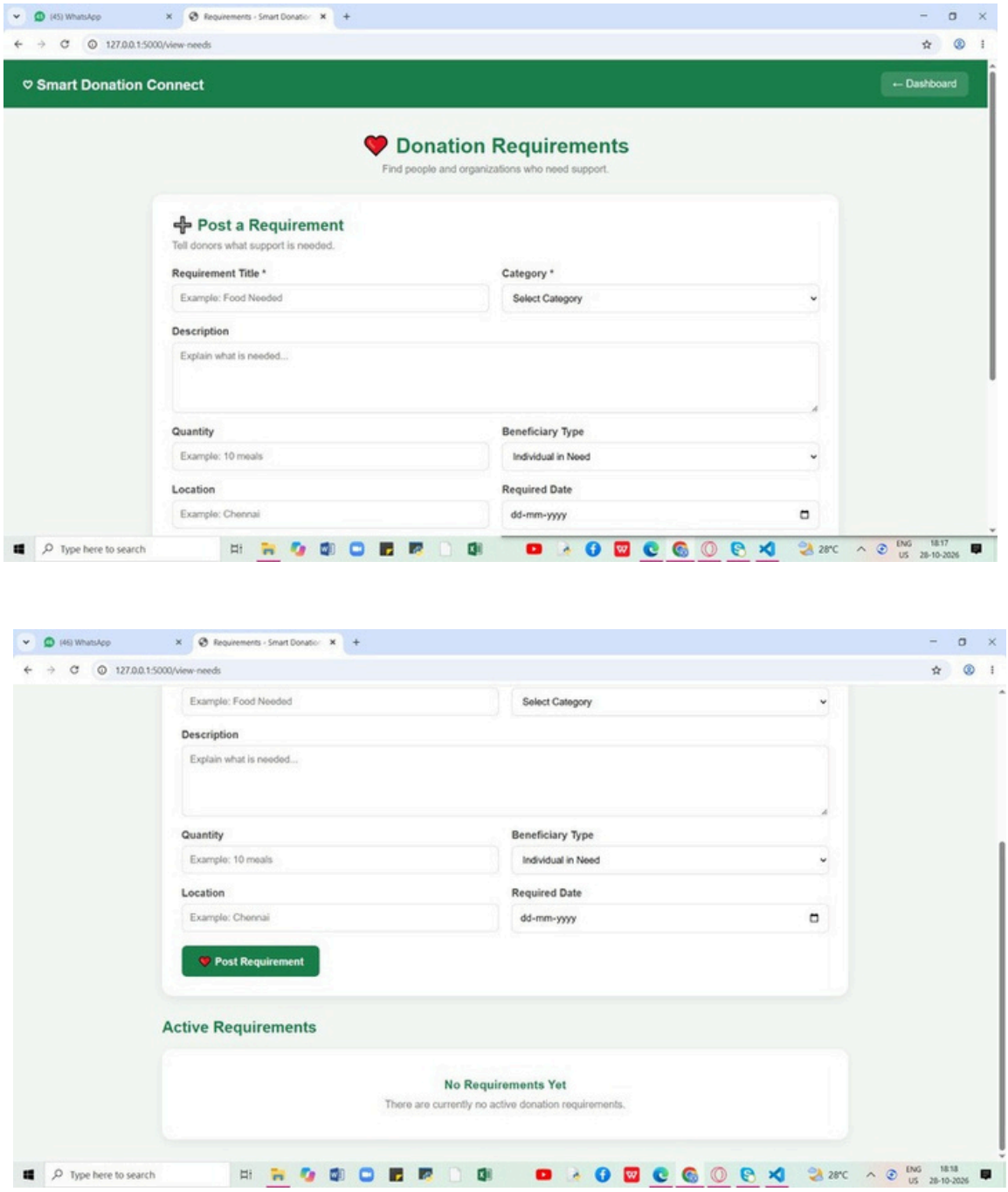


Figure-6 Requirement dashboard

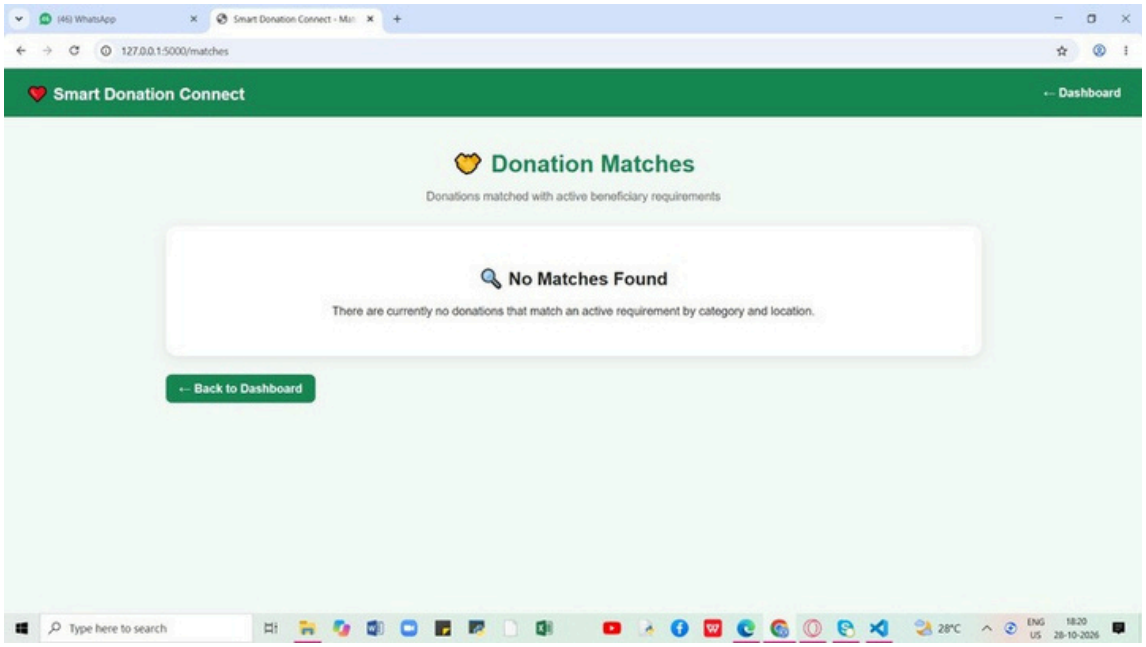


Figure-7 Matches dashboard

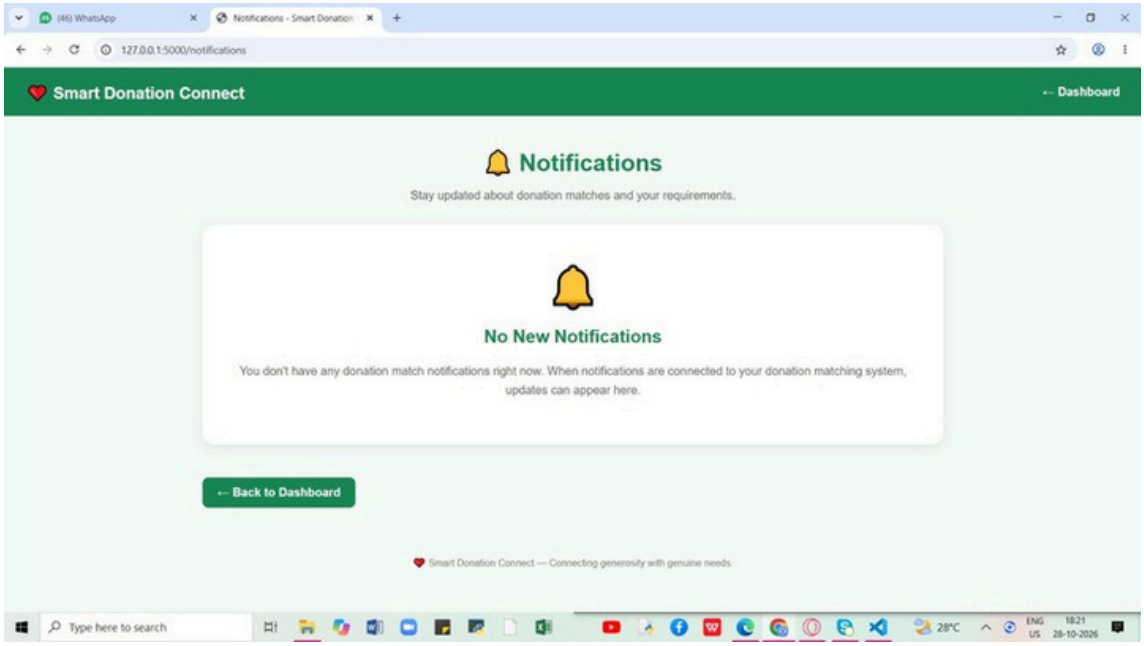


Figure -8 Notification dashboard

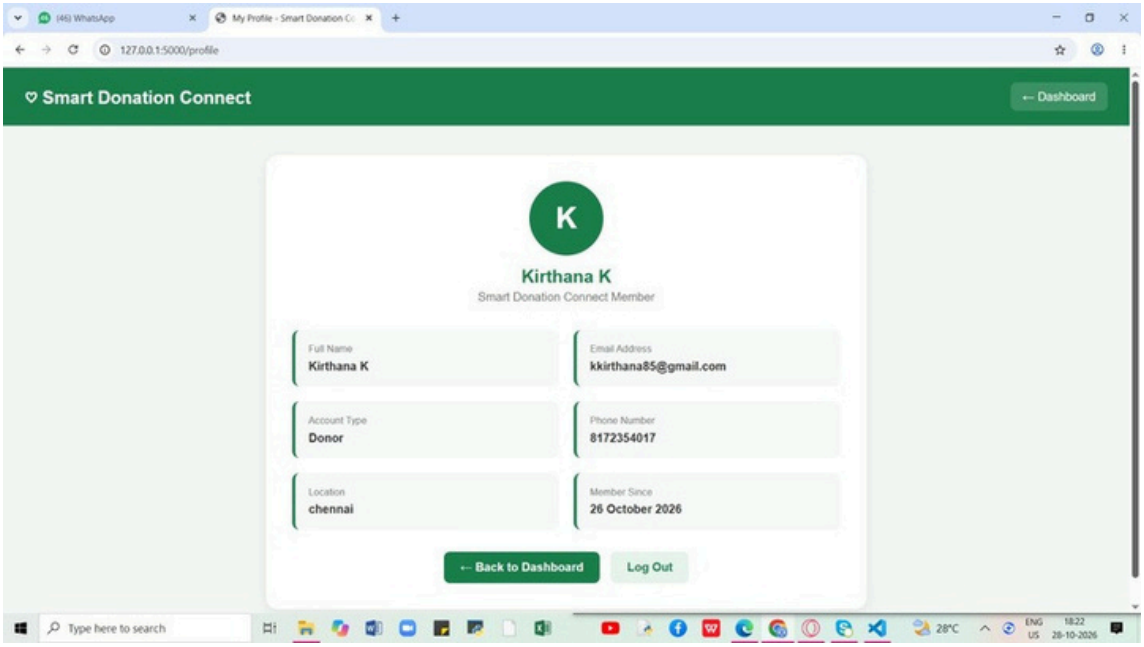
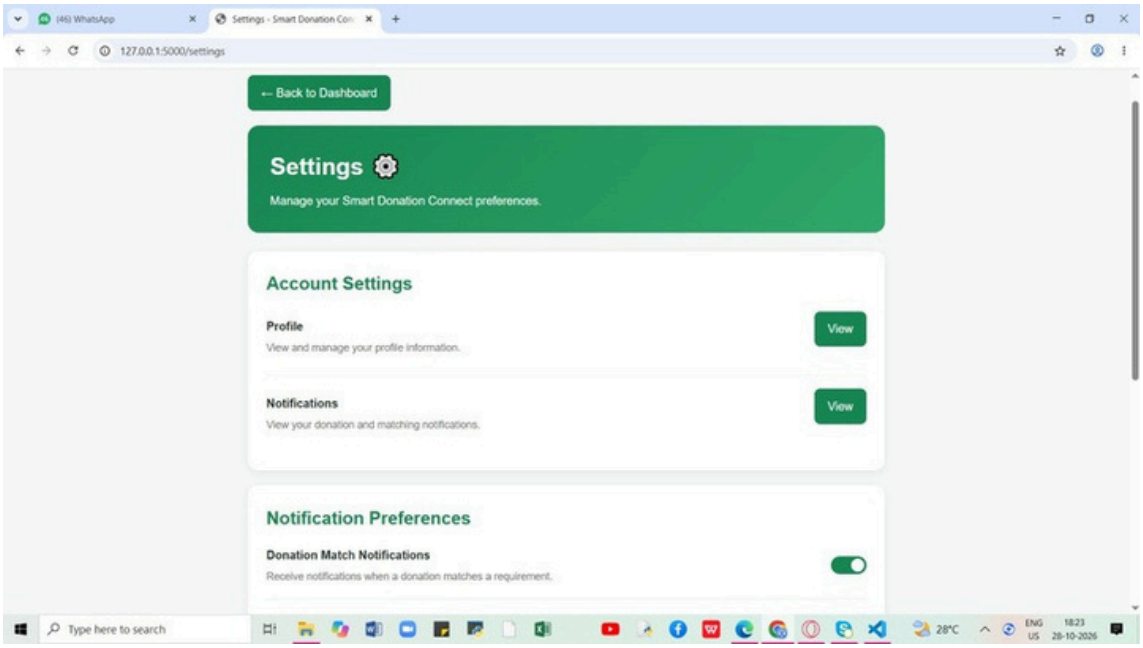


Figure-9 profile dashboard



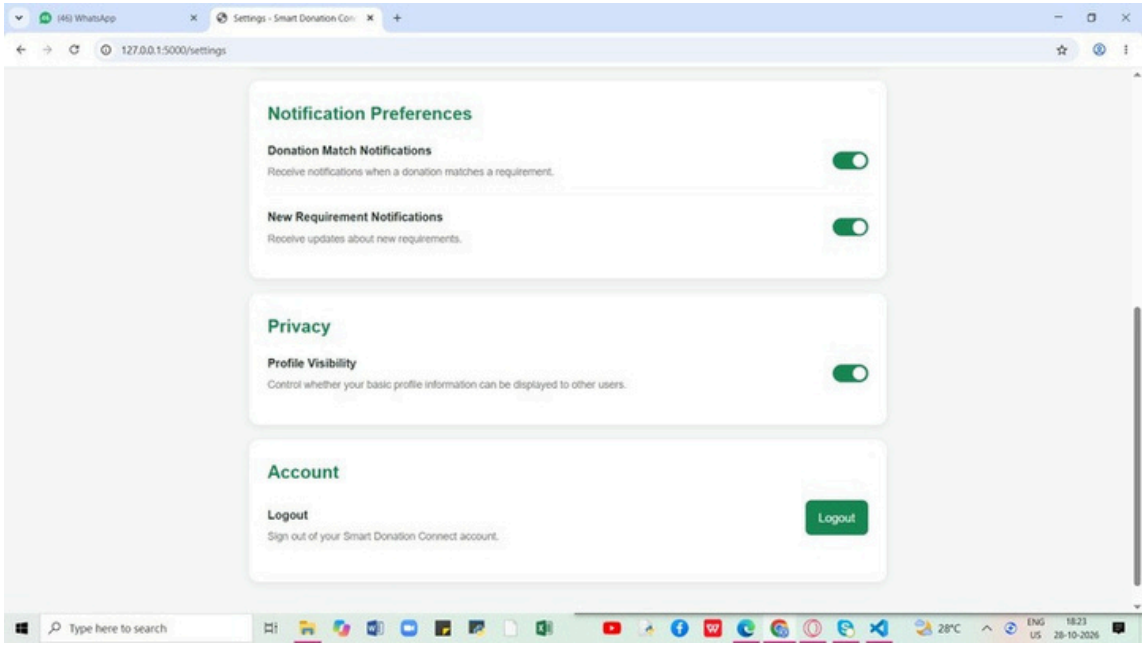


Figure-10 setting dashboard

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